

Shoprite-X Polling App - Administrator Guide

Shoprite-X Polling App — System Administrator Guide

Overview

As a system administrator, you are responsible for deploying, configuring, and maintaining the Shoprite-X Polling App. This guide covers setup, environment configuration, database management, and operational tasks.

System Architecture

The application consists of:

- **Frontend & API:** Next.js 14+ (App Router) deployed on Vercel
 - **Database:** Supabase Postgres (managed PostgreSQL)
 - **Storage:** Supabase Storage (for background images)
 - **Authentication:** Shared secret token (Phase 1)
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Deployment

Prerequisites

- Node.js 18+ installed
- A Supabase project (free tier is sufficient)
- A Vercel account (for deployment)
- Git repository (for CI/CD)

Initial Setup

1. **Clone the repository:**

```
git clone https://github.com/KuduBul/shoprite-x-polling-app.git
cd shoprite-x-polling-app
```

2. **Install dependencies:**

```
npm install
```

3. **Configure environment variables** (see section below)

4. **Set up the database:**

- Go to Supabase Dashboard → SQL Editor
- Run `prisma/init.sql` to create tables
- Optionally run `prisma/seed.sql` for sample data

5. **Generate Prisma client:**

```
npx prisma generate
```

6. **Start development server:**

```
npm run dev
```

7. **Deploy to Vercel:**

```
vercel deploy --prod
```

Environment Variables

Create a `.env` file in the project root:

Variable	Required	Default	Description
DATABASE_URL	Yes	—	Supabase Postgres connection string (pooler, port 6543)
DIRECT_URL	Yes	—	Direct Supabase connection (port 5432, for migrations)
ADMIN_SECRET	Yes	—	Shared secret for admin API authentication
RETENTION_RESPONSES_DAYS	No	90	Days to retain poll responses before purge
RETENTION_AUDIT_DAYS	No	365	Days to retain audit logs before purge
POLL_INTERVAL_MS	No	3000	Client polling interval for live results (ms)
SUPABASE_URL	Yes	—	Supabase project URL
SUPABASE_SERVICE_ROLE_KEY	Yes	—	Supabase service role key (for storage)
NEXT_PUBLIC_SUPABASE_URL	Yes	—	Public Supabase URL (for client-side)
NEXT_PUBLIC_SUPABASE_ANON_KEY	Yes	—	Public anon key (for client-side)

Connection String Format

For IPv6-only Supabase projects (newer projects):

```
DATABASE_URL="postgresql://postgres.[project-ref]:[password]@aws-1-[region].pooler.supabase.com:6543/postgres?pgbouncer=true"
DIRECT_URL="postgresql://postgres.[project-ref]:[password]@aws-1-[region].pooler.supabase.com:5432/postgres"
```

Security Notes

- Use a strong, unique ADMIN_SECRET (minimum 16 characters recommended)
- Never commit .env to version control
- Rotate the admin secret periodically
- The service role key has full database access — keep it server-side only

Database Management

Schema

The database has 4 tables:

Table	Purpose
Poll	Poll metadata, facilitator state, soft-delete flag
Question	Questions with options, positions, display order
Response	Participant responses with deduplication constraint
AuditLog	Immutable action log for governance

Running Migrations

```
npx prisma db push      # Push schema changes to database
npx prisma generate     # Regenerate client after schema changes
```

Backup

Supabase provides automatic daily backups on paid plans. For free tier: - Use pg_dump via the Supabase CLI - Or export data via the Supabase Dashboard

Data Retention & Purge

The system automatically purges old data:

Data Type	Retention Period	Configurable Via
Poll responses	90 days	RETENTION_RESPONSES_DAYS
Audit logs	365 days	RETENTION_AUDIT_DAYS
Soft-deleted polls	30 days (hard-delete)	Not configurable

Automated Purge

- Configured via Vercel Cron (vercel.json)
- Runs daily at **02:00 UTC**

- Endpoint: POST /api/system/purge
- Logs a DATA_PURGED audit entry with counts

Manual Purge

curl -X POST https://your-app.vercel.app/api/system/purge

Rate Limiting

- **Limit:** 60 requests per minute per IP address
- **Scope:** All public endpoints (participant-facing)
- **Response:** HTTP 429 with RATE_LIMITED error code
- **Implementation:** In-memory sliding window (resets on server restart)

Audit Logging

All significant actions are logged immutably:

Action	Trigger
POLL_CREATED	Admin creates a poll
POLL_UPDATED	Admin updates poll metadata
POLL_DELETED	Admin soft-deletes a poll
POLL_CLONED	Admin clones a poll
QUESTION_CREATED	Admin adds a question
QUESTION_UPDATED	Admin modifies a question
QUESTION_DELETED	Admin removes a question
RESPONSES_SUBMITTED	Participant submits responses
RESPONSES_RESET	Admin resets poll responses
FACILITATOR_SESSION_STARTED	Facilitator opens dashboard
FACILITATOR_STATE_UPDATED	Any facilitator control change
REVEAL_STAGE_CHANGED	Reveal stage toggled
VOTING_OPENED / VOTING_CLOSED	Voting toggled
DATA_PURGED	Automated purge executed

Querying Audit Logs

Via Supabase SQL Editor:

```
SELECT * FROM "AuditLog"
WHERE "pollId" = 'your-poll-id'
ORDER BY "createdAt" DESC
LIMIT 50;
```

Monitoring & Troubleshooting

Health Checks

- The app responds to any route — a 200 on / confirms it’s running
- Check Vercel deployment logs for server errors
- Check Supabase Dashboard → Logs for database issues

Common Issues

Issue	Cause	Solution
401 on all admin routes	Wrong ADMIN_SECRET	Verify env var matches what users enter
Database connection timeout	Wrong connection string	Check DATABASE_URL format and region
“Tenant or user not found”	Wrong pooler region	Verify the aws-x-[region] in your URL
Images not uploading	Missing storage config	Check SUPABASE_SERVICE_ROLE_KEY
Cron not running	Vercel config issue	Verify vercel.json cron configuration

Performance

- Designed for **100 concurrent participants** per poll
 - API responses target **<500ms at p95**
 - Database connection pool: **10 connections**
 - Client polling interval: **3 seconds**
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Scaling Considerations

Current Limitations (Phase 1)

- In-memory rate limiter resets on deploy/restart
- No WebSocket support (polling-based updates)
- Single admin secret (no per-user accounts)
- No horizontal scaling for rate limiting

Future Phases

- **Phase 2:** Per-user authentication, persistent rate limiting
 - **Phase 3:** Supabase Realtime for instant updates, WebSocket support
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Security Checklist

- ☐ Strong ADMIN_SECRET configured (16+ characters)
 - ☐ .env file excluded from version control
 - ☐ Supabase RLS policies reviewed (currently disabled for app access)
 - ☐ Service role key only used server-side
 - ☐ Rate limiting active on public endpoints
 - ☐ XSS sanitisation active on free-text inputs
 - ☐ HTTPS enforced (automatic on Vercel)
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Shoprite-X Polling App v0.1.0